Use-Case Flow Specification - UC-03: Reserve Tool

Community Tool & Equipment Library | Problem Statement #52 | Media, Events & Community

Primary Actor: Community Member
Secondary Actor: Payment Gateway

Preconditions

- The member is registered and logged in (UC-01).
- The tool exists in the inventory with status "Available" in a depot locker.
- The member has a payment method on file for the deposit hold.

Postconditions

- Success: a reservation with a unique ID, pickup window and due date is stored; the deposit hold is placed; the tool status is "Reserved"; the member has a locker pickup code.
- Failure: no reservation or deposit hold exists and the tool's availability is unchanged.

Main Success Scenario

1 Member searches the inventory (UC-02) and selects a tool, e.g. "Cordless Drill".
2 System shows the tool's category, depot locker, live availability, maximum loan duration and deposit amount.
3 Member chooses a pickup date and loan duration.
4 System checks that the duration does not exceed the category limit.
5 System verifies member eligibility (UC-06, «include»): no overdue items, no unpaid fees, fewer than 3 active loans.
6 System calculates the deposit (UC-07, «include») and shows a summary: deposit, due date and late fee per day.
7 Member accepts the loan terms and confirms the reservation.
8 System requests a deposit hold from the Payment Gateway.
9 Payment Gateway authorizes the hold and returns a payment token.
10 System saves the reservation, marks the tool "Reserved" for a 24-hour pickup window and generates a locker pickup code.
11 System sends a confirmation (email and SMS) with the pickup code, locker location and due date.
12 Use case ends successfully.

Alternate Flow

5a. Member is not eligible (overdue item or unpaid fees)

5a1. System shows "Borrowing suspended" and lists the overdue items and fees owed.
5a2. System offers a link to View Loans & Due Dates (UC-04) to resolve them.
5a3. Use case ends; no reservation or deposit hold is created.